APPTITUDE NOTES FOR INTERVIEW PREP.

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1. Main Categories & Topics

I. Quantitative Aptitude (Quants)

1. Numbers & Operations:

A. HCF & LCM

1. Find the HCF and LCM of 72 and 120.
2. Two numbers are in the ratio 3:4, and their LCM is 84. What is the smaller number?
3. Calculate the greatest number that divides 201 and 845 leaving remainders 9 & 5, respectively.
4. The product of two numbers is 180, and their HCF is 3. Find their LCM.
5. If the LCM of two numbers is 168 and their HCF is 12, and one of the numbers is 24, find the other number.

B. Number Systems

1. Determine the number of trailing zeros in 100!.
2. Is 153 a prime number? Explain.
3. Find the sum of the first 50 natural numbers.
4. If a number leaves a remainder of 2 when divided by 5, what will be the remainder when the square of this number is divided by 5?
5. Prove that the sum of any three consecutive integers is divisible by 3.

C. Decimals & Fractions

1. Convert 0.75 into a fraction.
2. Simplify: 3.5 + 2.75 - 1.25.
3. If 2/3 of a number is 12, what is the number?
4. Arrange in ascending order: 0.5, 2/3, 0.75, 1/3.
5. Multiply 0.25 by 0.4 and express the result as a fraction.

D. Surds & Indices

1. Simplify: $\sqrt{50} + \sqrt{18}$.
2. Express 81 as a power of 3.
3. Simplify: (2^3) × (2^4).
4. If 5^(2x) = 125, find the value of x.
5. Rationalize the denominator: 5 / $\sqrt{2}$.

E. Divisibility Rules

1. Is 234 divisible by 9?
2. Determine if 123456 is divisible by 11.
3. Check if 357 is divisible by 3 and 5.
4. Is 1024 divisible by 8?
5. Find the smallest number divisible by both 6 and 9.

F. Ages

1. A father is twice as old as his son. Five years ago, the sum of their ages was 60. Find their current ages.
2. The average age of a group of 5 people is 30 years. If a new person joins the group, the average age becomes 32 years. Find the age of the new person.
3. In 10 years, a mother will be twice as old as her daughter. Currently, the mother is 28 years old. How old is the daughter?
4. The ratio of the ages of two friends is 3:4. In 5 years, the ratio will be 4:5. Find their current ages.
5. A man is 24 years older than his son. In two years, his age will be twice that of his son's age. What are their current ages?

G. Inverse Problems

1. If 5 workers can complete a task in 20 days, how long will it take 10 workers to complete the same task?
2. A car travels 60 km in 1.5 hours. What is its average speed?
3. If 8 machines can produce 400 units in 5 hours, how many units can 5 machines produce in 8 hours?
4. A pipe can fill a tank in 3 hours, while another can empty it in 6 hours. If both are opened simultaneously, how long will it take to fill the tank?
5. If 12 men can build a wall in 15 days, how many men are required to build the same wall in 10 days?

2. Speed, Time & Work

A. Speed, Time & Distance

1. A car travels 150 km in 3 hours. What is its average speed?
2. If a person walks at a speed of 5 km/h for 2 hours and then at 4 km/h for 3 hours, what is the total distance covered?
3. Two trains start from the same point and travel in opposite directions at speeds of 60 km/h and 80 km/h, respectively. How far apart will they be after 2 hours?
4. A cyclist covers a certain distance in 2 hours at a speed of 15 km/h. How long will it take to cover the same distance at a speed of 10 km/h?
5. If a train 200 meters long passes a pole in 20 seconds, what is its speed in km/h?

B. Work & Time Problems

1. If A can complete a task in 10 days and B can do it in 15 days, how long will it take them to complete it together?
2. Three workers can complete a job in 6 days. If one worker leaves after 2 days, how long will it take the remaining workers to finish the job?
3. A and B together can complete a work in 8 days. B alone can complete it in 12 days. How long will A alone take to complete the work?
4. If 5 men can paint a house in 20 days, how many men are required to paint it in 10 days?
5. A machine can produce 100 units in 5 hours. How many units can it produce in 8 hours?

C. Boats & Streams

1. A boat can travel at 13 km/h in still water. If the speed of the stream is 4 km/h, how long will it take the boat to go 68 km downstream?
2. A woman can row upstream at 6 km/h and downstream at 12 km/h. What is the speed of the stream?
3. A boat running downstream covers a distance of 16 km in 2 hours, while for covering the same distance upstream, it takes 4 hours. What is the speed of the boat in still water?
4. A boat's speed in still water is 10 km/h, and the speed of the stream is 2 km/h. How long will it take the boat to travel 30 km downstream and 30 km upstream?
5. In one hour, a boat goes 11 km along the stream and 5 km against the stream. What is the speed of the boat in still water?

D. Pipes & Cisterns

1. Pipe A can fill a tank in 20 hours, and Pipe B can fill it in 30 hours. If both pipes are opened together, how long will it take to fill the tank?
2. Pipe A can fill a tank in 12 hours, while Pipe B can empty it in 18 hours. If both pipes are opened together, how long will it take to fill the tank?
3. Three pipes A, B, and C can fill a tank in 15 hours, 20 hours, and 30 hours, respectively. If all three are opened together, how long will it take to fill the tank?
4. Pipe A can fill a tank in 6 hours, and Pipe B can fill it in 8 hours. However, Pipe C can empty the tank in 12 hours. If all three pipes are opened together, how long will it take to fill the tank?
5. A cistern has two inlet pipes that can fill it in 4 hours and 6 hours, respectively, and an outlet pipe that can empty it in 8 hours. If all three pipes are opened together, how long will it take to fill the cistern?

3. Banking & Commercial Mathematics:

A. Averages, Ratio & Proportion

Averages

1. The average of 5 numbers is 60. If one number is excluded, the average becomes 55. What is the excluded number?
2. The average of three numbers is 42. Two of the numbers are 38 and 48. Find the third number.
3. The average of 10 consecutive even numbers is 48. Find the smallest number.
4. The average score of a batsman in 5 matches is 45. If the scores in 4 matches are 40, 50, 55, and 35, what is the score in the fifth match?
5. The average height of 8 boys is 162 cm. Two new boys with heights of 168 cm and 174 cm join the group. Find the new average height.

Ratio & Proportion:

1. Divide ₹1200 between A and B in the ratio 3:5.
2. If A and B are in the ratio 5:7 and B and C are in the ratio 2:3, find the ratio between A and C.
3. Two numbers are in the ratio 4:5. If their sum is 72, find the two numbers.
4. A sum of ₹3000 is divided between A, B, and C in the ratio 2:3:5. How much will each person receive?
5. If the ratio of the ages of A and B is 3:4 and the sum of their ages is 56, find their present ages

B. Simple & Compound Interest

Simple Interest:

1. Find the simple interest on ₹5000 at a rate of 5% per annum for 3 years.
2. At what rate of interest will ₹8000 amount to ₹9600 in 2 years under simple interest?
3. A sum of ₹12000 gives an interest of ₹1800 in 3 years. Find the rate of interest per annum.
4. Find the principal if the simple interest for 5 years at 8% per annum is ₹4000.
5. What is the simple interest on ₹10000 at 6% per annum for 2.5 years?

Compound Interest:

1. Find the compound interest on ₹6000 at 5% per annum for 2 years.
2. A sum of ₹5000 amounts to ₹5832 in 2 years at compound interest. Find the rate of interest.
3. At what rate of compound interest per annum will a sum of ₹4000 become ₹4840 in 2 years?
4. A sum of ₹8000 amounts to ₹8820 in 2 years at compound interest. Find the rate of interest.
5. The compound interest on ₹2500 at 4% per annum for 2 years is ₹204. Find the principal.

C. Profit & Loss, Percentages

Profit & Loss:

1. A shopkeeper buys a shirt for ₹800 and sells it for ₹1000. Find the profit percentage.
2. An article is sold for ₹540 at a loss of 10%. Find the cost price.
3. A man sells an article at 20% profit. If the cost price is ₹500, find the selling price.
4. A shopkeeper marks an article 20% above the cost price and gives a discount of 10%. Find the profit or loss percentage.
5. An article is bought for ₹1200 and sold for ₹1500. Find the profit percentage.

Percentages:

1. What is 25% of 600?
2. A student's score increased from 60 to 75. What is the percentage increase?
3. Find the percentage change when a value decreases from 80 to 64.
4. If 40% of a number is 120, find the number.
5. A salary increases by 10% and then decreases by 10%. Find the overall percentage change.

D. Allegations & Mixtures

1. A container has 40 liters of milk and 10 liters of water. How much water should be added to make the ratio of milk to water 2:1?
2. A 20-liter solution contains 8 liters of alcohol. How much water should be added to make it a 30% alcohol solution?
3. Two alloys of gold and silver are in the ratio 3:4 and 5:7, respectively. In what ratio should they be mixed to get an alloy with a gold-to-silver ratio of 2:3?
4. A container contains 60 liters of a mixture of milk and water in the ratio 3:2. How much water must be added to make the ratio 1:1?
5. Two types of rice, one costing ₹50 per kg and another costing ₹60 per kg, are mixed in a ratio of 2:3. What is the cost of the mixture per kg?

E. Successive Discounts

1. An article is marked at ₹1200. Successive discounts of 10% and 5% are given. Find the selling price.
2. Find the net discount when successive discounts of 20% and 10% are applied to an item.
3. A product is marked at ₹2000. A shopkeeper gives two successive discounts of 15% and 5%. Find the final selling price.
4. If a dress is marked at ₹1500 and a shopkeeper gives three successive discounts of 10%, 5%, and 2%, find the final price.
5. A mobile phone is marked at ₹18000. It is sold at two successive discounts of 20% and 10%. Find the selling price.

4.Progressions

A. Arithmetic Progression (AP)

1. Find the 10th term of an AP where the first term a=5a = 5a=5 and the common difference d=3d = 3d=3.
2. The 7th term of an AP is 20, and the 15th term is 40. Find the first term and the common difference.
3. If the sum of the first 15 terms of an AP is 120, and the first term is 2, find the common difference.
4. Determine the number of terms in the AP: 3, 7, 11, ..., up to 99.
5. The sum of three numbers in AP is 27, and their product is 504. Find the numbers.

B. Geometric Progression (GP)

1. Find the 5th term of a GP where the first term a=2a = 2a=2 and the common ratio r=3r = 3r=3.
2. The 3rd term of a GP is 24, and the 6th term is 192. Find the first term and the common ratio.
3. If the sum of the first 4 terms of a GP is 30, and the first term is 2, find the common ratio.
4. Determine the sum to infinity for the GP: 8, 4, 2, ...
5. The product of three numbers in GP is 512, and their sum is 26. Find the numbers.

C. Harmonic Progression (HP)

1. If three numbers are in HP, and the first and third numbers are 4 and 12, respectively, find the second number.
2. Determine the nth term of an HP whose corresponding AP has the first term 2 and common difference 3.
3. The sum of three numbers in HP is 15, and their product is 80. Find the numbers.
4. If the reciprocals of three numbers are in AP, and the numbers are 1, x, and 1/3, find x.
5. Find the harmonic mean of 5 and 20.

5.Probability & Combinatorics

A. Basic Probability

1. A fair coin is tossed three times. What is the probability of getting exactly two heads?
2. In a standard deck of 52 cards, what is the probability of drawing an Ace or a King?
3. A box contains 5 red balls and 3 blue balls. Two balls are drawn at random without replacement. What is the probability that both balls are red?
4. If two dice are rolled simultaneously, what is the probability that the sum of the numbers is 7?
5. A bag contains 4 white, 3 black, and 2 red balls. One ball is drawn at random. What is the probability that it is either black or red?

B. Permutations & Combinations (including Circular Permutations)

1. How many ways can the letters of the word "PROBABILITY" be arranged?
2. In how many ways can a committee of 3 men and 2 women be formed from a group of 5 men and 4 women?
3. How many different 5-digit numbers can be formed using the digits 1, 2, 3, 4, and 5 without repetition?
4. In how many ways can 6 people be seated around a circular table?
5. From a group of 10 people, in how many ways can a president, a vice-president, and a secretary be chosen?

6.Geometry & Algebra

A. Basic Geometry (Perimeter, Area, Volume)

1. Calculate the area and perimeter of a rectangle with a length of 8 cm and a width of 5 cm.
2. Find the volume of a cylinder with a radius of 3 cm and a height of 10 cm.
3. Determine the area of a triangle with a base of 6 cm and a height of 4 cm..
4. What is the perimeter of a regular hexagon with each side measuring 7 cm?
5. Calculate the surface area of a sphere with a radius of 5 cm.

B. Heights & Distances

1. A tree casts a shadow of 15 meters when the angle of elevation of the sun is 30 degrees. Find the height of the tree.
2. From the top of a 50-meter-high building, the angle of depression to a car on the ground is 45 degrees. How far is the car from the base of the building?
3. Two buildings are 100 meters apart. From the top of the shorter building (50 meters high), the angle of elevation to the top of the taller building is 30 degrees. Find the height of the taller building.
4. An observer on a hill 120 meters high observes two objects on the ground on opposite sides of the hill. The angles of depression to the objects are 30 degrees and 45 degrees, respectively. Find the distance between the two objects.
5. A kite is flying at a height of 60 meters, and the string makes an angle of 40 degrees with the horizontal. Find the length of the string.

C. Coordinate Geometry, Venn Diagrams, Set Theory

1. Find the distance between the points (3, 4) and (7, 1) in the coordinate plane.
2. Determine the equation of the line passing through the points (2, 3) and (5, 7).
3. If sets A and B have 15 and 20 elements respectively, and their intersection has 5 elements, how many elements are in their union?
4. Draw a Venn diagram to represent the relationship between three sets A, B, and C, where $A \cap B \cap C \neq \emptyset$.
5. Solve for x and y: $x + y = 10$ and $x - y = 2$.

D. Linear & Quadratic Equations

1. Solve the linear equation: $3x - 5 = 16$.
2. Find the roots of the quadratic equation: $x^2 - 4x - 5 = 0$.
3. Determine the value of k for which the quadratic equation $x^2 + kx + 9 = 0$ has equal roots.
4. Solve the system of linear equations: $2x + 3y = 12$ and $x - y = 3$.
5. If one root of the quadratic equation $x^2 + px + 6 = 0$ is 2, find the value of p and the other root.

E. Logarithms

1. Solve for x: $\log_2(x) = 5$.
2. Simplify the expression: $\log_3(27) - \log_3(3)$.
3. If $\log_{10}(2) = 0.3010$ and $\log_{10}(3) = 0.4771$, find $\log_{10}(6)$.
4. Solve for x: $\log_{10}(x - 2) + \log_{10}(x + 2) = 1$.
5. Express $\log_{10}(1000)$ in terms of common logarithms.

7.Miscellaneous

A. Clocks & Calendars

Clocks:

1. At what time between 3:00 and 4:00 will the minute and hour hands of a clock be exactly overlapping?
2. Calculate the angle between the hour and minute hands at 5:20.
3. If a clock shows 8:30, what is the angle between the hour and minute hands?
4. How many times do the hour and minute hands of a clock overlap in a 12-hour period?
5. Determine the time between 2:00 and 3:00 when the angle between the hour and minute hands is exactly 90 degrees.

Calendars:

1. If January 1st, 2023, is a Sunday, what day of the week will October 31st, 2023, be?
2. Find the day of the week on August 15, 1947.
3. How many leap years are there between 2000 and 2100?
4. If today is Wednesday, what day will it be 45 days from now?
5. Determine the day of the week on July 4, 1776.

B. Remainder Problems

1. What is the remainder when 123456 is divided by 7?
2. Find the remainder when 2502^{50}250 is divided by 13.
3. If a number leaves a remainder of 5 when divided by 12, what will be the remainder when the same number is divided by 4?
4. Determine the remainder when the product of the first 10 natural numbers is divided by 11.
5. What is the remainder when 71037^{103}7103 is divided by 5?

C. Simplification and Approximation

1. Simplify: (25×4)+(36÷6)5\frac{(25 \times 4) + (36 \div 6)}{5}5(25×4)+(36÷6)
2. Approximate the value of 50\sqrt{50}50 to two decimal places.
3. Simplify: 3.5×2.8−1.4×0.53.5 \times 2.8 - 1.4 \times 0.53.5×2.8−1.4×0.5
4. Estimate the value of 198×202200\frac{198 \times 202}{200}200198×202 without using a calculator.
5. Simplify the expression: (152−92)6\frac{(15^2 - 9^2)}{6}6(152−92)

D. Trains and Races

1. Two trains, each 120 meters long, are moving in opposite directions at speeds of 60 km/h and 90 km/h. How long will it take for the trains to pass each other completely?
2. A train 150 meters long passes a pole in 15 seconds. What is the speed of the train in km/h?
3. 5In a 100-meter race, A beats B by 10 meters. If A's speed is 10 m/s, what is B's speed?
4. Two trains start from stations A and B, 200 km apart, and travel towards each other at speeds of 50 km/h and 70 km/h, respectively. How long will it take for them to meet?
5. A train passes a platform 100 meters long in 25 seconds and a man standing on the platform in 15 seconds. What is the length of the train?

E. Data Interpretation

1. Given a bar graph showing the sales of a company over five years, determine the year with the highest sales growth.
2. Analyze a pie chart representing the market share of different companies in a sector and identify which company holds the largest share.
3. Interpret a line graph depicting the population growth of a city over a decade and calculate the average annual growth rate.
4. From a table displaying the monthly expenditures of a household, determine the category with the highest average expenditure over six months.
5. Examine a histogram showing the distribution of test scores among students and identify the score range with the highest frequency.

1. Main Categories & Topics

II. Logical Reasoning

1. Series & Patterns

A. Number Series

1. Identify the next number in the series: 2, 6, 12, 20, 30, ?
2. Find the missing number: 5, 9, 17, 33, ?
3. Determine the next term in the sequence: 1, 4, 9, 16, 25, ?
4. What comes next in the series: 3, 8, 15, 24, 35, ?
5. Identify the missing number: 7, 14, 28, 56, ?

B. Letter & Symbol Series

1. Find the next term in the series: A, C, F, J, O, ?
2. What comes next: Z, X, U, Q, ?
3. Identify the missing term: B2D, D4F, F6H, ?
4. Determine the next term: M, P, S, V, ?
5. What comes next in the series: @, #, @, #, @, ?

C. Logical Sequences

1. Arrange the following words in a meaningful sequence: Seed, Plant, Tree, Fruit, Flower.
2. Which number logically follows in the sequence: 2, 4, 8, 16, ?
3. Identify the next figure in the sequence: [Image of a pattern sequence].
4. Find the next term: 5A, 10B, 15C, 20D, ?
5. Determine the missing number: 1, 1, 2, 6, 24, ?

D. Analogy & Classification

1. Dog is to Bark as Cat is to ?
2. Find the odd one out: Circle, Square, Triangle, Sphere.
3. Hand is to Glove as Foot is to ?
4. Which word does not belong: Apple, Orange, Banana, Carrot.
5. Book is to Reading as Knife is to ?

2. Statements & Assumptions

A. Drawing Conclusions from Statements

- Statement: All roses are flowers. Some flowers fade quickly.
  Conclusion: Some roses fade quickly.
  Is the conclusion correct?

- Statement: No dogs are cats. All cats are animals.
  Conclusion: No dogs are animals.
  Is the conclusion correct?

- Statement: Some students are intelligent. Intelligent people are successful.
  Conclusion: Some students are successful.
  Is the conclusion correct?

- Statement: All engineers are creative. Some creative people are artists.
  Conclusion: Some engineers are artists.
  Is the conclusion correct?

- Statement: No fruits are vegetables. All apples are fruits.
  Conclusion: No apples are vegetables.
  Is the conclusion correct?

B. Data Sufficiency

Question: What is the value of xxx?
Statements:

1. $2x+3=11$
2. $x-2=3$

Is the information sufficient?

Question: Is number nnn even?
Statements:

1. n is divisible by 4.
2. n is a multiple of 2.

Is the information sufficient?

Question: What is the area of rectangle RRR?
Statements:

1. The length of RRR is 10 units.
2. The perimeter of RRR is 30 units.

Is the information sufficient?

Question: Is y greater than 10?
Statements:

1. y is an integer.
2. $y^2>100y^2 > 100y^2>100$

Is the information sufficient?

Question: What is the value of a+ba + ba+b?
Statements:

1. $a=5$
2. $b=7$

Is the information sufficient?

C. Statement and Arguments

- Statement: Should the government implement a universal basic income?
  Arguments:
  a. Yes, it will reduce poverty and provide financial security.
  b. No, it will discourage people from working.
  Evaluate the arguments.

- Statement: Should public transportation be free for all citizens?
  Arguments:
  a. Yes, it will reduce traffic congestion and pollution.
  b. No, it will increase the financial burden on the government.
  Evaluate the arguments.

- Statement: Should schools adopt a four-day workweek?
  Arguments:
  a. Yes, it will improve student and teacher well-being.
  b. No, it will reduce instructional time and affect learning outcomes.
  Evaluate the arguments.

- Statement: Should the voting age be lowered to 16?
  Arguments:
  a. Yes, it will encourage political engagement among youth.
  b. No, 16-year-olds lack the maturity to make informed decisions.
  Evaluate the arguments.

- Statement: Should the voting age be lowered to 16?
  Arguments:
  a. Yes, it is unethical and causes unnecessary suffering.
  b. No, it is essential for medical research and drug development.
  Evaluate the arguments.

D. Statement and Conclusions

- Statement: All birds can fly. Penguins are birds.
  Conclusion: Penguins can fly.
  Is the conclusion valid?

- Statement: Some fruits are sweet. Mangoes are fruits.
  Conclusion: Mangoes are sweet.
  Is the conclusion valid?

- Statement: No mammals lay eggs. Platypuses are mammals.
  Conclusion: Platypuses do not lay eggs.
  Is the conclusion valid?

- Statement: All engineers are logical thinkers. Some logical thinkers are artists.
  Conclusion: Some engineers are artists.
  Is the conclusion valid?

- Statement: Some politicians are honest. All honest people are trustworthy.
  Conclusion: Some politicians are trustworthy.
  Is the conclusion valid?

E. Course of Action

- Problem: A city is facing severe water shortages during summer months.
  Course of Action:
  1. Implement water rationing and promote water conservation practices.
  2. Construct new dams and reservoirs to increase water storage capacity.

  Evaluate the courses of action.

- Problem: Increasing traffic accidents at a busy intersection.
  Course of Action:
  1. Install traffic lights and deploy traffic police to control the flow of vehicles.
  2. Introduce stricter penalties for traffic violations in the area.

  Evaluate the courses of action.

- Problem: High dropout rates in government schools.
  Course of Action:
  1. Introduce scholarship programs and financial incentives for students.
  2. Provide additional support, including counseling and free meals, to encourage students to continue their education.

  Evaluate the courses of action.

- Problem: The city is experiencing a rise in air pollution levels.
  Course of Action:
  1. Implement stricter regulations on industrial emissions and vehicle pollution.
  2. Encourage the use of public transport and electric vehicles through incentives.

  Evaluate the courses of action.

- Problem: A company is facing low employee morale and high turnover.
  Course of Action:
  1. Conduct employee engagement surveys and address the issues raised.
  2. Introduce performance-based incentives and improve working conditions.

  Evaluate the courses of action.

3.Visual & Spatial Reasoning

A. Cube, Cuboid & Dice Problems

1. A cube is painted on all six faces and then cut into 64 smaller equal cubes. How many of these smaller cubes will have exactly two painted faces?
2. A standard six-faced die is rolled. What is the probability that the number on the top face is greater than 4?
3. If two opposite faces of a cube are painted red and the remaining faces are painted blue, and then the cube is cut into 27 smaller equal cubes, how many of these smaller cubes will have at least one red face?
4. A cuboid has dimensions 4 cm × 3 cm × 2 cm. How many smaller cubes of side 1 cm can be formed from this cuboid?
5. A cube is divided into 27 smaller equal cubes, and then two opposite faces are removed. How many smaller cubes remain?

B. Mirror Image

1. Which of the following options represents the correct mirror image of the given figure? (Provide a figure and multiple-choice options)
2. Identify the mirror image of the word "LOGIC" when the mirror is placed vertically to the right of the word.
3. Determine the mirror image of the clock showing 3:45 when reflected in a vertical mirror.
4. Which option shows the correct mirror image of the given arrangement of shapes? (Provide an arrangement and multiple-choice options)
5. Find the mirror image of the number "12345" when the mirror is placed horizontally below the number.

C. Paper Folding

1. A square sheet of paper is folded twice along its diagonals and then a small circle is cut out from the center. When the paper is unfolded, how many circular holes will be present?
2. A rectangular paper is folded in half vertically and three triangular cuts are made along the folded edge. When unfolded, how many triangular holes will be visible?
3. If a paper is folded in half twice, first vertically and then horizontally, and a single hole is punched at the center, how many holes will be there when the paper is unfolded?
4. A circular paper is folded into four equal parts, and a square is cut out from the center. How many square holes will appear when the paper is unfolded?
5. A triangular sheet of paper is folded along its median, and a small circle is cut near the vertex. Upon unfolding, how many circular holes will be observed?

4.Classic Reasoning

A. Direction Sense Test

1. A person walks 5 km north, then turns right and walks 3 km, then turns right again and walks 5 km. In which direction is he now from the starting point?
2. If you are facing east and turn 90 degrees to your left, which direction are you facing now?
3. A person starts from point A and walks 10 km east to point B, then 5 km south to point C, then 10 km west to point D. What is the distance and direction of point D from point A?
4. John walks 7 km north, then 4 km east, then 7 km south. How far and in which direction is he from the starting point?
5. If you are facing south and turn 45 degrees to your right, which direction are you facing now?

B. Blood Relations

1. Pointing to a photograph, a woman says, "This man's sister is my mother's only daughter." How is the man related to the woman's mother?
2. If A is the brother of B, B is the sister of C, and C is the father of D, how is A related to D?
3. Introducing a boy, a girl says, "He is the son of the daughter of my father's mother." How is the boy related to the girl?
4. If P is the mother of Q and R, and S is the father of R, how is Q related to S?
5. Pointing to a man, Sarah says, "His mother is the only daughter of my mother." How is Sarah related to the man?

C. Odd One Out

1. Find the odd one out: Apple, Banana, Cherry, Carrot.
2. Find the odd one out: Circle, Square, Triangle, Sphere.
3. Find the odd one out: Dog, Cat, Rabbit, Eagle.
4. Find the odd one out: Red, Blue, Green, Circle.
5. Find the odd one out: Gold, Silver, Iron, Wood.

D. Syllogism

1. Statements: All cats are animals. Some animals are dogs. Conclusions: (i) Some cats are dogs. (ii) Some dogs are animals. Determine which conclusion(s) logically follow.
2. Statements: No fruits are vegetables. All apples are fruits. Conclusions: (i) No apples are vegetables. (ii) Some vegetables are fruits. Determine which conclusion(s) logically follow.

3. Statements: Some birds are mammals. All mammals are warm-blooded. Conclusions: (i) Some birds are warm-blooded. (ii) No birds are warm-blooded. Determine which conclusion(s) logically follow.
4. Statements: All roses are flowers. Some flowers are red. Conclusions: (i) Some roses are red. (ii) All flowers are roses. Determine which conclusion(s) logically follow.
5. Statements: Some students are intelligent. All intelligent people are hardworking. Conclusions: (i) Some students are hardworking. (ii) All hardworking people are intelligent. Determine which conclusion(s) logically follow.

E. Seating Arrangements

1. Five friends are sitting in a row facing north. A is to the immediate right of B, who is at one end. C is between D and E. Who is sitting in the middle?
2. Six people are sitting around a circular table. A is to the left of B, who is directly opposite C. D is between E and F. Who is sitting directly opposite D?
3. Eight people are sitting in two parallel rows containing four people each. In row 1, A, B, C, and D are seated and all of them are facing south. In row 2, P, Q, R, and S are seated and all of them are facing north. Each person in row 1 sits exactly opposite to a person in row 2. B is facing Q. A is at one end of the row. Who is facing P?
4. Seven people are sitting in a straight line facing north. P is fourth to the left of Q. R is between P and S. T is at one end. Who is sitting in the middle?
5. Four couples are sitting around a circular table. No couple sits together. If A is sitting to the immediate left of B, and B's spouse is C, who is sitting to the immediate right of C?

F. Puzzles

1. A farmer has 17 sheep and all but 9 die. How many are left?
2. You have a 3-liter jug and a 5-liter jug, and you need to measure exactly 4 liters of water. How can you do it?
3. A man is looking at a photograph of someone. His friend asks who it is. The man replies, "Brothers and sisters, I have none. But that man's father is my father's son." Who is in the photograph?
4. There are three boxes, one containing only apples, one containing only oranges, and one containing both apples and oranges. The boxes are labeled incorrectly. You can only open one box and take out one piece of fruit. How can you label all the boxes correctly?
5. A clock shows the time as 3:15. What is the angle between the hour and the minute hands?

5.Miscellaneous Logical Puzzles

A. Cryptarithmetic

- Solve the following cryptarithm:

```
  SEND
+ MORE
------------
 MONEY
```

- In the cryptarithm below, find the digits corresponding to each letter:

```
  TWO
+ TWO
---------
 FOUR
```

- Determine the digits for the following cryptarithm:

```
  BASE
+ BALL
---------
  GAME
```

- Solve this cryptarithm:

```
  CROSS
+ ROAD
---------
 DANGER
```

- Find the digits that satisfy the following cryptarithm:

```
  APPLE
+ APPLE
---------
 ORANGE
```

Note: Cryptarithmetic puzzles, also known as alphametics, involve replacing letters with digits to make a valid arithmetic equation. Each letter represents a unique digit, and leading digits cannot be zero.

B. Objective Reasoning

- If all Bloops are Razzies and all Razzies are Lazzies, are all Bloops definitely Lazzies?
  A) Yes
  B) No
  C) Cannot be determined

- If some Smugs are Blips and no Blips are Gruds, can some Smugs be Gruds?
  A) Yes
  B) No
  C) Cannot be determined

- All Jacks are Jills. Some Jills are not Janes. Are all Janes definitely not Jacks?
  A) Yes
  B) No
  C) Cannot be determined

- If every Flib is a Flob, and no Flob is a Fleb, is it true that no Flib is a Fleb?
  A) Yes
  B) No
  C) Cannot be determined

- Some Frams are not Frims. All Frims are Frooms. Are some Frams definitely not Frooms?
  A) Yes
  B) No
  C) Cannot be determined

C. Selection Decision Tables

A company has the following hiring criteria:

Applicants must have at least 3 years of experience.
Applicants must possess a relevant certification or a degree.
If the applicant has more than 5 years of experience, the certification/degr
requirement is waived.

Based on this, determine which of the following applicants should be selected:

Applicant A: 4 years of experience, no certification/degree.
Applicant B: 2 years of experience, has a degree.
Applicant C: 6 years of experience, no certification/degree.
Applicant D: 3 years of experience, has a certification.

A loan approval process uses the following rules:

Applicant's credit score must be above 700.
Applicant's annual income must be at least $50,000.
If the credit score is above 750, the income requirement is reduced to $40,000.

Which of the following applicants qualify for the loan?
Applicant A: Credit score 720, income $45,000.
Applicant B: Credit score 760, income $42,000.
Applicant C: Credit score 680, income $55,000.
Applicant D: Credit score 740, income $50,000.

A university admission policy states:

Students must have a GPA of 3.5 or higher.
Students must have extracurricular involvement.
If the GPA is 3.8 or higher, extracurricular involvement is optional.

Identify which applicants are eligible for admission:
Applicant A: GPA 3.6, involved in extracurricular activities.
Applicant B: GPA 3.9, no extracurricular activities.
Applicant C: GPA 3.4, involved in extracurricular activities.
Applicant D: GPA 3.7, no extracurricular activities.

D. Inferred Meaning

1. "The river is dry and the crops are failing." What is the implied cause of this situation?
- A) Poor farming methods
- B) Lack of rainfall
- C) Overuse of water resources
- D) Pesticide overuse
2. "John arrived at the office late and was scolded by his manager." What can be inferred about John's workplace?☐
- A) His office has strict rules about punctuality.
- B) John's manager is unfair.
- C) John is a lazy employee.
- D) John’s office allows flexible timing.
3. "The flight was delayed due to poor visibility." What is the most likely reason for this delay?☐
- A) Mechanical failure
- B) Weather conditions
- C) Air traffic control error
- D) Fuel shortage
4. "The student submitted the assignment late and received a penalty." What can you infer from this statement?☐
- A) The teacher is strict about deadlines.
- B) The student didn't understand the assignment.
- C) The student was careless.
- D) The assignment was difficult.
5. "The shop closed early due to low customer turnout." What is the most probable reason for this?☐
- A) A festival was being celebrated.
- B) The shop owner was unwell.
- C) A major sporting event was taking place.
- D) The shop was undergoing renovation.

1. Main Categories & Topics

III. Verbal Ability

1.Grammar & Basics

A. Parts of Speech

- Identify the part of speech of the underlined word:
  She quickly finished her homework.
  A) Noun
  B) Verb
  C) Adjective
  D) Adverb

- Choose the correct part of speech for the word "beautiful" in the following sentence:
  The beautiful painting was admired by everyone.
  A) Noun
  B) Verb
  C) Adjective
  D) Adverb

- In the sentence below, what part of speech is the word "and"?
  She bought apples and oranges.
  A) Preposition
  B) Conjunction
  C) Interjection
  D) Adverb

- Determine the part of speech of the word "Wow" in the following sentence:
  Wow! That's an amazing performance.
  A) Noun
  B) Verb
  C) Interjection
  D) Adjective

- Identify the part of speech of the underlined word:
  They will arrive soon.
  A) Noun
  B) Verb
  C) Adverb
  D) Adjective

B.Tenses

- Choose the correct form of the verb to complete the sentence:
  By this time next year, she ___________ her degree.
  A) will have completed
  B) will complete
  C) completes
  D) completed

- Identify the tense used in the following sentence:
  They have been working on the project for two weeks.
  A) Present Perfect
  B) Past Continuous
  C) Present Perfect Continuous
  D) Past Perfect

- Select the appropriate tense for the sentence:
  When I arrived, they ___________ dinner.
  A) have
  B) are having
  C) were having
  D) had

- Fill in the blank with the correct tense:
  She ___________ to the gym every day.
  A) goes
  B) went
  C) going
  D) gone

- Choose the correct tense to complete the sentence:
  By the time we got to the cinema, the movie ___________.
  A) started
  B) had started
  C) has started
  D) starts

C. Articles

- Select the correct article to complete the sentence:
  She adopted ___________ European lifestyle.
  A) a
  B) an
  C) the
  D) no article

- Choose the appropriate article for the sentence:
  He is ___________ honest man.
  A) a
  B) an
  C) the
  D) no article

- Fill in the blank with the correct article:
  I have ___________ idea.
  A) a
  B) an
  C) the
  D) no article

- Identify the correct article to use:
  She is ___________ best student in the class.
  A) a
  B) an
  C) the
  D) no article

- Choose the correct article for the sentence:
  He wants to buy ___________ new car.
  A) a
  B) an
  C) the
  D) no article

D. Prepositions

- Choose the correct preposition to complete the sentence:
  She is interested ____________ learning new languages.
  - A) on
  - B) in
  - C) at
  - D) for

- Select the appropriate preposition:
  The cat is hiding ____________ the table.
  - A) under
  - B) over
  - C) between
  - D) among

- Fill in the blank with the correct preposition:
  He will arrive ____________ Monday.
  - A) in
  - B) on
  - C) at
  - D) by

- Identify the correct preposition to use:
  They have been living here ____________ 2010.
  - A) for
  - B) since
  - C) from
  - D) by

- Choose the correct preposition for the sentence:
  She is good ____________ mathematics.
  - A) at
  - B) in
  - C) on
  - D) with

E. Conjunctions

Choose the correct conjunction to complete the sentence:
He was tired ____________ he continued to work.

- A) and
- B) but
- C) so
- D) because

Select the appropriate conjunction:
I'll call you ____________ I get home.

- A) and
- B) or
- C) when
- D) so

Fill in the blank with the correct conjunction:
Do you prefer tea ____________ coffee?

- A) but
- B) and
- C) or
- D) yet

Identify the conjunction in the following sentence:
She was feeling cold, ____________ she wore a sweater.

- A) and
- B) but
- C) so
- D) or

Choose the correct conjunction:
I like both pizza ____________ pasta.

- A) and
- B) or
- C) but
- D) yet

G. Subject-Verb Agreement

Choose the correct form of the verb:
The book and the pen ____________ *on the table.*

- A) is
- B) are
- C) was
- D) has

Identify the correct form of the verb:
Neither the teacher nor the students ____________ *ready.*

- A) is
- B) are
- C) was
- D) have

Select the correct verb for the sentence:
One of the boys ____________ *going to the market.*

- A) is
- B) are
- C) were
- D) have

Choose the correct verb:
The dog as well as the cats ____________ *sleeping.*

- A) is
- B) are
- C) have
- D) were

Identify the correct form of the verb:
Each of the students ____________ *a project.*

- A) has
- B) have
- C) were
- D) are

F. Idioms and Phrases

Choose the correct conjunction to complete the sentence:
He was tired ____________ he continued to work.

- A) and
- B) but
- C) so
- D) because

Select the appropriate conjunction:
I'll call you ____________ I get home.

- A) and
- B) or
- C) when
- D) so

Fill in the blank with the correct conjunction:
Do you prefer tea ____________ coffee?

- A) but
- B) and
- C) or
- D) yet

Identify the conjunction in the following sentence:
She was feeling cold, ____________ she wore a sweater.

- A) and
- B) but
- C) so
- D) or

Choose the correct conjunction:
I like both pizza ____________ pasta.

- A) and
- B) or
- C) but
- D) yet

2. Sentence Skills:

A. Sentence Ordering

Definition: Sentence ordering involves arranging a set of jumbled sentences into a coherent and logical paragraph.

Example:

Given the following sentences:

A) He decided to go for a walk.
B) It started raining heavily.
C) He forgot to take his umbrella.
D) John was bored at home.

Correct Order:

D) John was bored at home. A) He decided to go for a walk. C) He forgot to take his umbrella. B) It started raining heavily.

Tips to Solve Sentence Ordering Questions:

1. Identify the Opening Sentence: Look for an introductory sentence that introduces the main idea or subject.
2. Find Logical Connections: Determine how sentences relate to each other, focusing on chronological order, cause and effect, or general to specific information.
3. Use Transition Words: Words like "however," "therefore," and "meanwhile" can provide clues about the sequence.
4. Determine the Closing Sentence: The concluding sentence often summarizes the paragraph or presents a final thought.

B. Jumbled Sentences

Definition: Jumbled sentences are exercises where the words of a sentence or the sentences of a paragraph are presented in a mixed order, and the task is to rearrange them to form a meaningful sentence or paragraph.

Example:
Jumbled Words: "to / I / market / the / went"

Correct Sentence: "I went to the market."

Strategies to Solve Jumbled Sentences:

1. Identify the Subject: Determine who or what the sentence is about.
2. Find the Verb: Locate the action or state of being.
3. Look for Objects and Complements: Identify what is receiving the action or additional information about the subject.
4. Arrange in SVO Order: Most English sentences follow the Subject-Verb-Object structure.

C. Error Identification

Definition: Error identification involves spotting grammatical, punctuation, or usage errors in sentences.

Example:
"She don't like ice cream."

Error: "don't" should be "doesn't."

Tips for Error Identification:

1. Subject-Verb Agreement: Ensure subjects and verbs agree in number.
2. Tense Consistency: Check that verb tenses are consistent throughout the sentence.
3. Pronoun Reference: Verify that pronouns clearly refer to the correct nouns.
4. Punctuation: Look for correct use of commas, periods, and other punctuation marks.

D. Sentence Improvement & Construction

Definition: This involves enhancing a sentence by correcting grammatical errors, improving clarity, or restructuring it for better flow.

Example:
Original: "Despite of the rain, we went outside."

Improved: "Despite the rain, we went outside."

Strategies for Sentence Improvement:
1. Eliminate Redundancies: Remove unnecessary words that don't add meaning.
2. Use Active Voice: Prefer active voice over passive for clarity.
3. Simplify Complex Sentences: Break down overly complex sentences into simpler ones.
4. Ensure Parallelism: Maintain the same grammatical structure in lists or comparisons.

3.Paragraph & Comprehension:

A. Cloze Test

Definition: A Cloze Test is an exercise where words are removed from a passage at regular intervals, and the participant is required to fill in the blanks. This assesses the reader's understanding of context and vocabulary.

Example:
"The quick brown ____ jumps over the lazy ____."
Answer: "fox," "dog"

Tips to Approach Cloze Tests:

1. Read the Entire Passage First: Understand the overall context before filling in the blanks.
2. Use Context Clues: Look at the words and sentences surrounding the blank to infer the correct word.
3. Consider Grammar and Syntax: Ensure the word fits grammatically within the sentence.
4. Think About Collocations: Some words commonly pair together; recognizing these can help.

B. Fill-in-the-Blanks

Definition: Similar to Cloze Tests, Fill-in-the-Blanks exercises involve completing sentences or passages by inserting appropriate words, focusing on vocabulary, grammar, or specific content knowledge.

Example:
"She ____ to the market to buy some vegetables."
Answer: "went"

Strategies:

- Identify the Part of Speech Needed: Determine if the blank requires a noun, verb, adjective, etc.
- Use Semantic Cues: The meaning of the sentence can guide you to the correct word.

C. Paragraph Ordering

Definition: This involves arranging a set of jumbled sentences into a coherent and logical paragraph, assessing understanding of logical flow and coherence.

Example:
1. "He opened his umbrella."
2. "It started to rain."
3. "John was walking to the store."

Correct Order:
1. "John was walking to the store."
2. "It started to rain."
3. "He opened his umbrella."

Approach:
- Identify the Topic Sentence: Look for the sentence that introduces the main idea.
- Determine Logical Sequence: Arrange sentences based on cause and effect, time order, or general to specific information.
- Look for Transition Words: Words like "however," "then," or "therefore" can provide clues to the correct order.

D. Reading Comprehension

Definition: Reading Comprehension involves understanding, interpreting, and analyzing written texts. It assesses the ability to grasp main ideas, details, inferences, and the author's intent.

Strategies:
- Skim the Passage First: Get a general idea of the content.
- Read Questions Before Detailed Reading: Knowing the questions can help focus on relevant information.
- Highlight or Note Key Points: Mark important information or sections related to the questions.
- Infer Meaning: Read between the lines to understand implied information.

E. Contextual Vocabulary

Definition: This refers to understanding the meaning of words based on the context in which they appear, rather than relying solely on dictionary definitions.

Example:

"The arid climate made farming difficult."

Inference: "Arid" means dry or lacking moisture.

Techniques to Enhance Contextual Vocabulary:

- Use Context Clues: Look for synonyms, antonyms, explanations, or examples in the surrounding text.
- Break Down the Word: Analyze prefixes, suffixes, and root words.
- Practice Regularly: Engage with diverse reading materials to encounter words in different contexts.

4. Synonyms & Antonyms:

A. Synonyms

Definition:

A synonym is a word that has a similar meaning to another word. It helps in enhancing vocabulary and improving the ability to use varied language while maintaining the same idea.

Example:

- Happy → Joyful, Cheerful, Content
- Fast → Quick, Rapid, Speedy

✅ Sample Questions for Synonyms

1. Select the synonym of the word "Abundant":
    - A) Rare
    - B) Scarce
    - C) Plentiful
    - D) Little
    - Answer: C) Plentiful
2. Choose the synonym of "Genuine":
    - A) Fake
    - B) Authentic
    - C) Counterfeit
    - D) Fraudulent
    - Answer: B) Authentic
3. Find the synonym of "Diverse":
    - A) Uniform
    - B) Mixed
    - C) Similar
    - D) Constant
    - Answer: B) Mixed
4. Select the synonym of "Innovative":
    - A) Traditional
    - B) Old-fashioned
    - C) Creative
    - D) Routine
    - Answer: C) Creative

B. Antonyms

Definition:
An antonym is a word that has the opposite meaning of another word.
Example:

- Happy → Sad, Unhappy, Miserable
- Fast → Slow, Sluggish

✅ Sample Questions for Antonyms

1. Select the antonym of the word "Optimistic":
    - A) Cheerful
    - B) Hopeful
    - C) Pessimistic
    - D) Joyful
    - Answer: C) Pessimistic
2. Choose the antonym of "Adversity":
    - A) Challenge
    - B) Problem
    - C) Opportunity
    - D) Hardship
    - Answer: C) Opportunity
3. Find the antonym of "Mandatory":
    - A) Compulsory
    - B) Optional
    - C) Obligatory
    - D) Required
    - Answer: B) Optional
4. Identify the antonym of "Gregarious":
    - A) Friendly
    - B) Outgoing
    - C) Introverted
    - D) Social
    - Answer: C) Introverted
5. Select the antonym of "Vulnerable":
    - A) Weak
    - B) Defenseless
    - C) Strong
    - D) Open
    - Answer: C) Strong

5. Miscellaneous:

A. Spellings

Definition:
Spelling questions test the ability to identify correctly spelled words or detect spelling errors in a sentence.

✅ Sample Questions for Spellings

1. Choose the correctly spelled word:
    - A) Accomodate
    - B) Accommodate
    - C) Acommodate
    - D) Accommadate
    - Answer: B) Accommodate
2. Identify the correct spelling:
    - A) Consciencious
    - B) Consciensious
    - C) Conscientious
    - D) Consientious
    - Answer: C) Conscientious
3. Choose the word with the correct spelling:
    - A) Recieve
    - B) Receive
    - C) Receeve
    - D) Receave
    - Answer: B) Receive
4. Find the correctly spelled word:
    - A) Beggining
    - B) Begining
    - C) Beginning
    - D) Begenning
    - Answer: C) Beginning
5. Pick the correctly spelled word:
    - A) Embarrass
    - B) Embarass
    - C) Embarras
    - D) Imbarass
    - Answer: A) Embarrass

B. One-Word Substitution

Definition:
One-word substitution questions require replacing a phrase or sentence with a single word.

✅ Sample Questions for One-Word Substitution

1. One who speaks many languages:
   - A) Linguist
   - B) Multilingual
   - C) Philologist
   - D) Orator
   - Answer: A) Linguist
2. A person who eats too much:
   - A) Cannibal
   - B) Glutton
   - C) Omnivore
   - D) Epicure
   - Answer: B) Glutton
3. One who does not believe in the existence of God:
   - A) Atheist
   - B) Agnostic
   - C) Skeptic
   - D) Believer
   - Answer: A) Atheist
4. A person who hates mankind:
   - A) Optimist
   - B) Misogynist
   - C) Misandrist
   - D) Misanthrope
   - Answer: D) Misanthrope
5. Government by a king or queen:
   - A) Democracy
   - B) Oligarchy
   - C) Monarchy
   - D) Theocracy
   - Answer: C) Monarchy

C. Active & Passive Voice

Definition:
Active voice emphasizes the subject performing the action, while passive voice emphasizes the action or object.

- Active Voice: John wrote the letter.
- Passive Voice: The letter was written by John.

✅ Sample Questions for Active & Passive Voice

- Convert to passive voice:
  She completed the project.
  - A) The project completes her.
  - B) The project was completed by her.
  - C) The project completes.
  - D) She was completed by the project.
  - Answer: B) The project was completed by her.
- Convert to active voice:
  The book was read by Sam.
  - A) Sam reads the book.
  - B) Sam is reading the book.
  - C) Sam read the book.
  - D) Sam will read the book.
  - Answer: C) Sam read the book.
- Convert to passive voice:
  The teacher is giving a lecture.
  - A) A lecture is being given by the teacher.
  - B) The lecture gives the teacher.
  - C) A lecture will be given by the teacher.
  - D) The teacher gives the lecture.
  - Answer: A) A lecture is being given by the teacher.
- Convert to active voice:
  The problem was solved by the engineer.
  - A) The engineer solves the problem.
  - B) The engineer solved the problem.
  - C) The problem is solved by the engineer.
  - D) The engineer is solving the problem.
  - Answer: B) The engineer solved the problem.

D. Direct & Indirect Speech

Definition:

- Direct Speech: Reporting the speaker's exact words using quotation marks.
- Indirect Speech: Reporting the idea without quoting the speaker's exact words.

✅ Sample Questions for Direct & Indirect Speech

1. Convert to indirect speech:
   He said, "I am going home."
   - A) He said he is going home.
   - B) He said that he was going home.
   - C) He said I am going home.
   - D) He said that I am going home.
   - Answer: B) He said that he was going home.
1. Convert to direct speech:
   She told me that she was tired.
   - A) She said, "I am tired."
   - B) She said, "She is tired."
   - C) She said, "You are tired."
   - D) She said, "I was tired."
   - Answer: A) She said, "I am tired."
1. Convert to indirect speech:
   He asked, "What is your name?"
   - A) He asked what was my name.
   - B) He asked what my name was.
   - C) He asked what is your name.
   - D) He asked what is my name.
   - Answer: B) He asked what my name was.
1. Convert to direct speech:
   She said that she was feeling cold.
   - A) She said, "I feel cold."
   - B) She said, "I am feeling cold."
   - C) She said, "She feels cold."
   - D) She said, "She felt cold."
   - Answer: B) She said, "I am feeling cold."
1. Convert to indirect speech:
   He said, "I have completed the task."
   - A) He said that he had completed the task.
   - B) He said that he completed the task.
   - C) He said that he was completing the task.
   - D) He said that he is completing the task.
   - Answer: A) He said that he had completed the task.

E. Verbal Analogies

Example:

Doctor : Hospital :: Teacher : ?

- A) Home
- B) School
- C) Office
- D) Library
- Answer: B) School

2.
Preparation Strategies & Tips

✅ **Build a Strong Foundation:**

• Revise basic mathematical concepts, grammar rules, and logical reasoning techniques.

✅ **Practice Regularly:**

• Solve practice tests and previous years' questions to build speed and accuracy.

✅ **Time Management:**

• Aptitude tests are usually timed; practice solving problems within set time limits.

✅ **Learn Shortcuts & Tricks:**

• Familiarize yourself with quick methods to solve common types of questions.

✅ **Utilize Online Resources:**

• Use websites like IndiaBIX, PrepInsta, and JobTestPrep for sample questions and tutorials.

✅ **Mock Tests:**

• Regularly attempt mock tests to simulate exam conditions and identify weak areas.

✅ **Stay Calm & Focused:**

• Practice stress management techniques to maintain a clear mind during the test.

3. Complete Aptitude Preparation Cheat Sheet

6. Complete Aptitude Preparation Cheat Sheet

I. Quantitative Aptitude

1. Numbers

Formula	Explanation
Sum of first n natural numbers	$S = n(n + 1) / 2$
Sum of squares of first n natural numbers	$S = n(n + 1)(2n + 1) / 6$
Sum of cubes of first n natural numbers	$S = (n(n + 1)/2)^2$
Sum of first n odd numbers	$S = n^2$
Sum of first n even numbers	$S = n(n + 1)$

2. HCF & LCM

Formula	Explanation
HCF × LCM	Product of the two numbers
HCF of fractions	HCF of numerators / LCM of denominators
LCM of fractions	LCM of numerators / HCF of denominators

3. Pipes and Cisterns

Formula	Explanation
Rate of filling/emptying	1/x per hour
Combined rate	$1/x + 1/y$
One fills, other empties	$1/x - 1/y$

4. Speed, Time, and Distance

Formula	Explanation
Speed	Distance ÷ Time
Distance	Speed × Time
Time	Distance ÷ Speed
Average Speed	(2xy) / (x + y)
Speed conversions	1 km/hr = 5/18 m/s; 1 m/s = 18/5 km/hr

II. Logical Reasoning

1. Number Series

Formula	Explanation
Arithmetic Series	Difference between consecutive terms is constant
Geometric Series	Ratio between consecutive terms is constant

2. Coding-Decoding

Formula	Explanation
Alphabet Shifting	Letters are shifted by a fixed number
Reverse Coding	Letters are reversed

3. Blood Relations

Formula	Explanation
Direct Relation	Father, Mother, Brother, Sister
Indirect Relation	Uncle, Aunt, Cousin

III. Verbal Ability

1. Spotting Errors

Concept	Explanation
Subject-Verb Agreement	The subject and verb must agree
Tense Consistency	Use the same tense throughout

2. Synonyms and Antonyms

Concept	Explanation
Synonyms	Words with similar meaning
Antonyms	Words with opposite meaning

3. Ordering of Sentences

Concept	Explanation
Logical Flow	Arrange sentences to form a meaningful paragraph
Chronological Order	Arrange sentences based on time sequence

4. Most Important Aptitude Topics for Placements

7. Most Important Aptitude Topics for Placements

1. HCF & LCM

- HCF × LCM = Product of two numbers
- HCF of fractions = HCF of numerators ÷ LCM of denominators
- LCM of fractions = LCM of numerators ÷ HCF of denominators

2. Speed, Time & Distance

- Speed = Distance ÷ Time
- Distance = Speed × Time
- Time = Distance ÷ Speed
- Average Speed = (2xy) ÷ (x + y)

3. Profit & Loss

- Profit = Selling Price – Cost Price
- Loss = Cost Price – Selling Price
- Profit % = (Profit ÷ Cost Price) × 100
- Loss % = (Loss ÷ Cost Price) × 100

4. Simple & Compound Interest

- SI = (P × R × T) ÷ 100
- CI = P(1 + R/100)^T - P

5. Probability

- Probability = Number of favorable outcomes ÷ Total outcomes
- Complementary Probability = 1 – Probability

6. Permutations & Combinations

- Permutation: P(n, r) = n! ÷ (n – r)!
- Combination: C(n, r) = n! ÷ (r!(n – r)!)

7. Syllogism

- Universal Affirmative: All A are B
- Universal Negative: No A is B
- Particular Affirmative: Some A are B

8. Blood Relations

- Father's son = Brother
- Son's wife = Daughter-in-law

9. Direction Sense

- Start facing North → Turn Right = East
- Start facing East → Turn Left = North

10. Statement & Conclusion

- Direct Conclusion = Based on facts stated
- Indirect Conclusion = Logical interpretation

11. Number Series

- Arithmetic Series = Constant difference
- Geometric Series = Constant ratio

12. Reading Comprehension

- Skim the passage
- Focus on the main idea

14. Coding-Decoding

- Letter shifting
- Reverse order

15. Data Interpretation

- Focus on patterns
- Break down data